# Recovering Deleted and Damaged Files (5e)

Digital Forensics, Investigation, and Response, Fifth Edition - Lab 03

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## Introduction

Data recovery techniques are a principal ancestor of modern digital forensics and have played a foundational role in its evolution. In the early days of computing, digital forensics was largely a matter of finding and recovering data from standalone hard drives. As technology has advanced and introduced widespread networking and connectivity, digital forensics has emerged from multiple disciplines, including data recovery, cybersecurity, network analysis, and traditional investigative methods. However, the ability to recover deleted or damaged files remains a cornerstone of forensic investigations.

While today's forensic investigations can span computers, networks, smartphones, and cloud storage, traditional data recovery is still a critical component of the field. Cybercriminals will still try to erase or corrupt data, so the ability to recover these files is still essential for uncovering illicit activities. On the other hand, data recovery can help individuals and corporations recover files from cyberattacks and assist in tracing cyberattack timelines.

When a user deletes a file, it’s usually moved to the Trash (macOS) or Recycle Bin (Windows OS), similar to placing something in a trash can at home. Just like you can easily pull out something from your trash, files in the Trash or Recycle Bin can easily be restored. A file will be considered "deleted" only after the user “empties” the Trash or Recycle Bin. However, this still does not actually remove the data. Instead, the system removes the reference to the file, like deleting the map that tells where the file was located. The file’s actual data remains on the hard drive until new data is written over it, which means that unless the storage space is reused, the file can often be recovered with special tools.

Of course, there are many other ways of deleting data that aren’t reversible. More secure and effective methods include delete with overwriting and physical destruction. A delete with overwriting means that the data is deleted and then new random data is written to the drive several times, making the recovery of the original files very difficult or impossible. Physical destruction, on the other hand, is just that—the destruction of the actual hard drive along with all the data stored on it. This can be accomplished through various methods, including shredding, melting, or crushing the hard drive, and ensures that the data cannot be recovered. Organizations handling highly sensitive information often employ physical destruction to ensure that the stored data is completely irretrievable.

The importance of data recovery cannot be overstated. In 2017, one of the three U.S. credit reporting agencies, Equifax, suffered one of the largest data breaches in U.S. history. The names, home addresses, phone numbers, birthdays, and social security numbers of over 148 million Americans were compromised in the breach. Equifax hired a security firm to investigate the breach, and forensic investigators were able to piece together everything that had happened (*Source: Electronic Privacy Information Center: Equifax data breach: https://archive.epic.org/privacy/data-breach/equifax/).*

In this lab, you will assume the role of a Digital Forensics Specialist who has been assigned to a case involving intellectual property theft. After obtaining a search warrant, the local authorities have seized multiple computers from the suspects and transferred images of their hard drives to the digital forensics team. So far, no incriminating evidence has been found on the drive images, but your boss, the lead investigator, believes the suspects may have deleted incriminating evidence. As a Digital Forensics Specialist, you will use professional data recovery tools and techniques to recover deleted data from different operating systems and file systems.

**Lab Overview**

**SECTION 1** of this lab consists of two parts, which should be completed in the order specified.

1. In the first part of the lab, you will recover data from a Windows NTFS drive image using E3.

2. In the second part of the lab, you will recover data from a Linux Ext4 drive image using Autopsy.

**SECTION 2** of this lab allows you to apply what you learned in SECTION 1 with less guidance.

1. In the first part of the lab, you will simulate data deletion and corruption.

1. In the second part of the lab, you will recover the deleted and corrupted data using data carving.

Finally, you will explore the virtual environment on your own in **SECTION 3** of this lab to complete challenges that allow you to use the skills you learned in the lab to conduct independent, unguided work—similar to what you will encounter in a real-world situation.

**Learning Objectives**

Upon completing this lab, you will be able to:

1. Understand how files are deleted and recovered.

2. Recover deleted files from NTFS, Ext4, FAT, and APFS drive images.

3. Recover deleted files using data carving techniques.

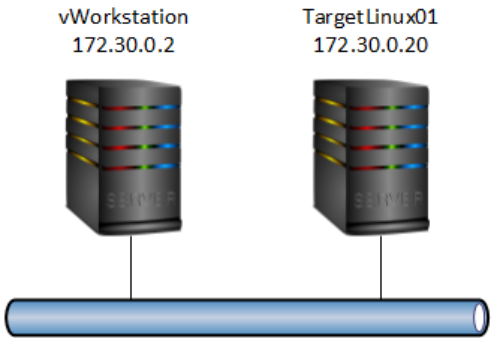
4. Recover deleted files from a live Windows system.

5. Recover deleted files from a live Linux system.

**Topology**

This lab environment consists of:

- vWorkstation (Windows: Server 2019)
- TargetLinux01 (Linux: Ubuntu)




**Tools and Software**

The following software and/or utilities are required to complete this lab. Students are encouraged to explore the Internet to learn more about the products and tools used in this lab.

- E3
- Autopsy
- PhotoRec

**Deliverables**

Upon completion of this lab, you are required to provide the following deliverables to your instructor:

**Hands-on Demonstration**

1. Lab Report file including screen captures of the following:

    - Recovered files and folders in the E3 Trash folder
    - Patent file within E3's File Viewer
    - Recovered files in File Explorer
    - List of deleted files in Autopsy
    - Recovered patent file in Adobe Reader

2. Any additional information as directed by the lab:

    - None

**Applied Learning**

1. Lab Report file including screen captures of the following:

    - Contents of the RAR archive in the /mnt/media/home/ash directory
    - Failed mount attempt on the /dev/sdb5 device
    - Compressed files recovered by PhotoRec
    - Backup files recovered from the extracted RAR archive

2. Any additional information as directed by the lab:

    - None

**Challenge and Analysis**

1. Lab Report file including screen captures of the following:

    - Patent file recovered from the FAT32 drive image within E3
    - Patent file recovered from the APFS drive image within Autopsy

2. Any additional information as directed by the lab:

    - None

## Hands-On Demonstration

**Note:** In this section of the lab, you will follow a step-by-step walk-through of the objectives for this lab to produce the expected deliverable(s).

1. **Review** the **Tutorial**.

Frequently performed tasks, such as making screen captures and downloading your Lab Report, are explained in the Cloud Lab Tutorial. The Cloud Lab Tutorial is available from the User menu in the upper-right corner of the Student Dashboard. You should review these tasks before starting the lab.

**Proceed** with **Part 1**.

**Part 1: Recover Deleted Files from NTFS Drive Image**

In this part of the lab, you will recover deleted files from a Windows drive image, a critical skill in digital forensic investigations. Windows remains the most widely used desktop operating system in both home and business environments, making it a frequent source of digital evidence.

The New Technology File System (NTFS) plays a significant role in forensic analysis due to its advanced metadata structures, journaling capabilities, and file recovery mechanisms. Features such as the Master File Table (MFT), transaction logs, and alternate data streams (ADS) provide valuable forensic artifacts that can assist in reconstructing user activity and recovering deleted files. However, these same features also present challenges, such as file system overwrites and anti-forensic techniques that investigators must navigate.

To analyze the NTFS-formatted drive image, you will use E3, a professional-grade digital forensic tool designed for in-depth evidence examination. E3 is particularly useful for this task due to its ability to efficiently parse NTFS metadata, recover deleted files, and handle forensic disk images in industry-standard formats. Its comprehensive search and indexing capabilities enable forensic analysts to uncover and interpret digital evidence with precision.

By completing this section, you will develop practical skills that are directly applicable to real-world forensic investigations. You will acquire and verify forensic disk images for data integrity, identify and parse NTFS metadata to track file deletions and modifications and analyze file system artifacts to reconstruct user activity. These skills are essential for digital forensic professionals working in law enforcement, corporate investigations, and cybersecurity incident response.

1. On the vWorkstation desktop, **double-click** the **Electronic Evidence Examiner icon** to open the E3 application.



E3 icon

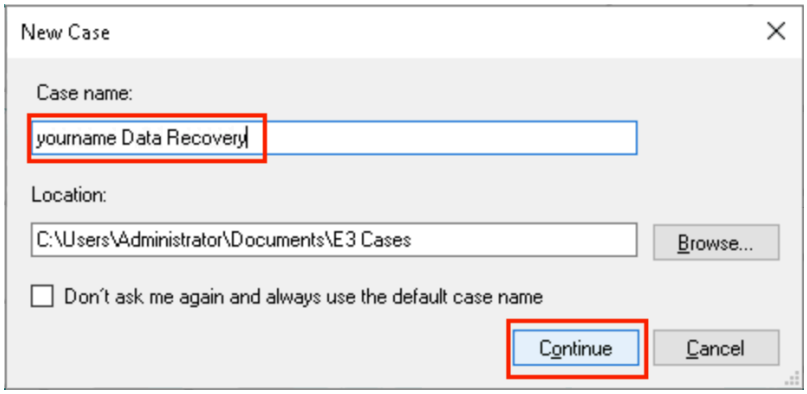
**Note:** E3 may take several minutes to load. The E3 welcome screen opens with shortcuts to the most common activities that forensic investigators will perform within the tool.

2. Close the opening screen, to view the Welcome screen. On the Welcome screen, **click** the **Add Evidence button** to open the New Case dialog box.



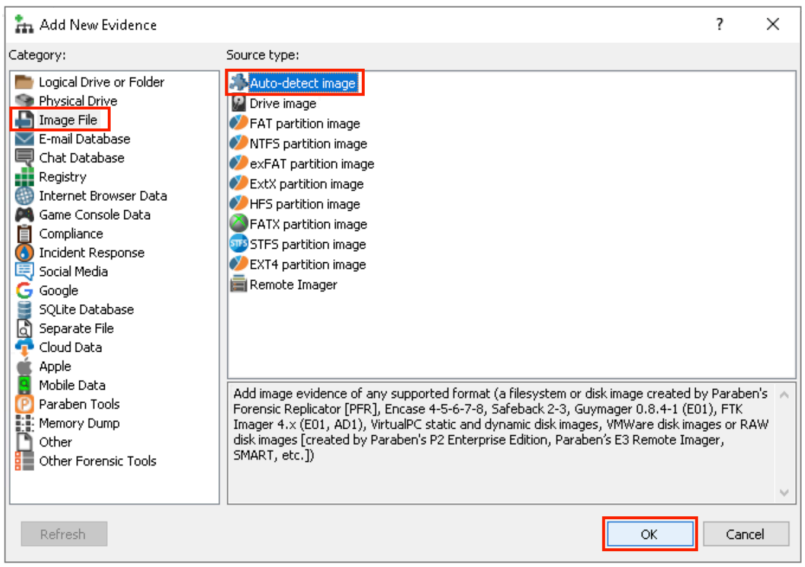

Welcome page - Add Evidence

3. In the New Case dialog box, **type** ***yourname*** **Data Recovery** in the Case name field, replacing *yourname* with your own name, then **click Continue** to create your new case file and open the Add New Evidence dialog box.




New Case dialog box

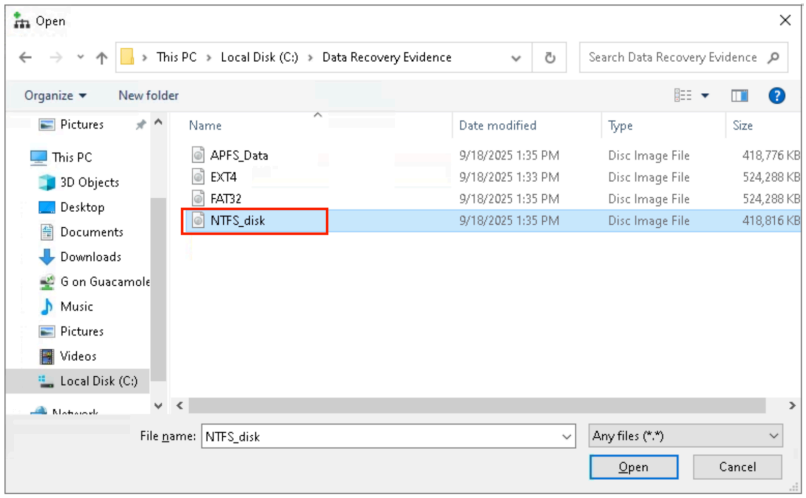
4. In the Add New Evidence dialog box, **click** the **Image File category**, then **select** the **Auto-detect image Source type** and **click OK** to continue.




Add New Evidence - Auto-detect image

**Note:** When importing drive images as evidenced in E3, you can manually define the type of file system used on the drive image if you know it. However, E3 can quickly determine this on its own using the Auto-detect image function.

5. In the Open dialog box, **navigate** to the **Data Recovery Evidence folder** (This PC > Local Disk (C:) > Data Recovery Evidence) and **double-click** the **NTFS_disk.img** to select the digital drive image for this lab.




Open dialog box

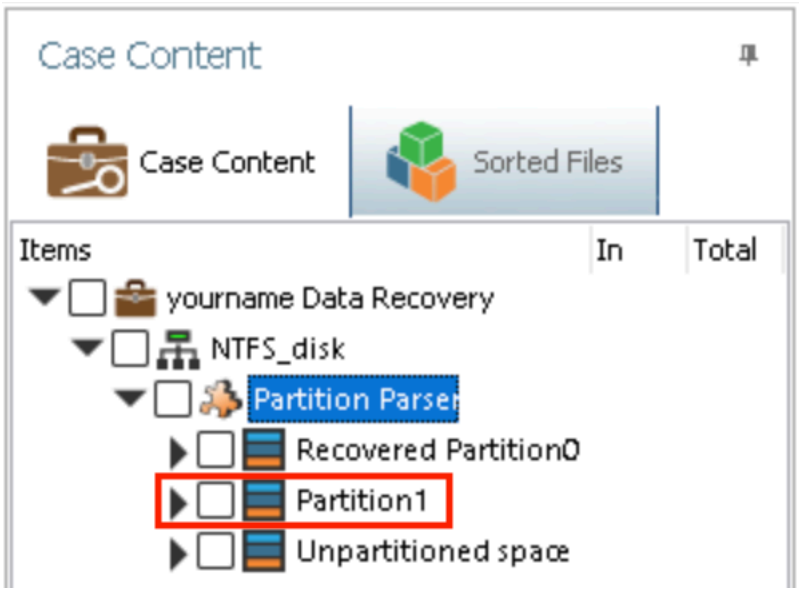
6. When prompted, **click OK** to accept the default name for the drive image and add the drive image to the Case Content pane.




Evidence name

**Note:** The yourname Data Recovery case will appear in the Case Content pane. The Case Content pane is the primary display and navigation area for all evidence in the open case file. Within the Case Content pane's tree-view structure, you can access case nodes, evidence nodes, evidence type nodes, folder nodes, and data grids/streams. Technically, the Case Content pane does not store the actual evidence but provides a series of links to elements within the physical evidence. When you select a specific node or grid within the Case Content pane, the contents of the enclosing folder or database will be displayed in the Data Viewer in the center pane. Within the Data Viewer, you can select individual files, folders, and records. Additional information about the currently selected node, grid, file, folder, or record will be displayed in the Viewers pane on the right, which provides multiple ways to view your current selection.

7. In the Case Content pane, **navigate** to ***yourname* Data Recovery\NTFS_disk\Partition Parser\Partition1** to open the NTFS Settings dialog box for Partition1.




Partition1

**Note:** E3 is capable of automatically restoring deleted files and folders when drive images are imported as evidence. However, for NTFS file systems, this process can take a great deal of time. For this reason, users are

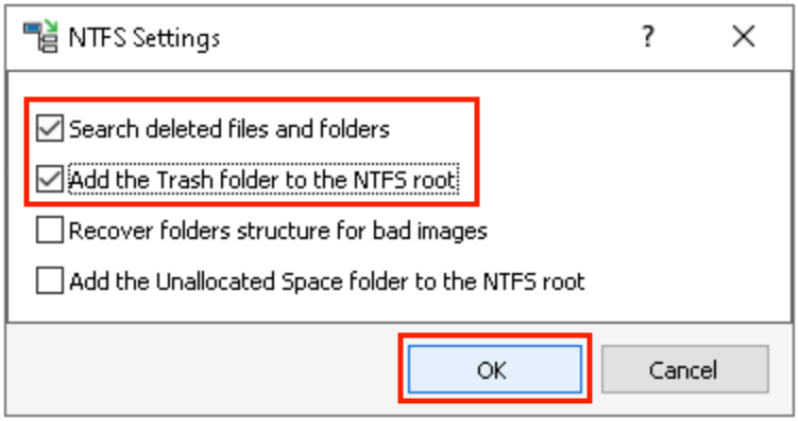
presented with the NTFS Settings dialog box when they access NTFS evidence. The NTFS Settings allow the user to determine if deleted data should be restored by E3, and if so, to what extent. When E3 and similar tools retrieve deleted data from NTFS file systems, they do so by referencing NTFS metafiles. In the context of digital forensics and data recovery, there are two critical metafiles in the NTFS file system: the Master File Table ($MFT) and the bitmap ($BitMap). The Master File Table serves as the file system's principal record of metadata for all files and folders in the system. This includes details like file name, location, creation date, permissions, size, and the clusters where the data is stored. The bitmap file is a record of which clusters on the disk are allocated and unallocated. When a file is deleted by emptying the Recycle Bin in an NTFS file system, the data is still not actually deleted. The corresponding record in the Master File Table is simply updated to indicate that that file has been deleted. The record will remain intact, as will the file data, at least until the corresponding clusters—now marked as unallocated space—are overwritten. Using this information, E3 and similar tools are able to quickly identify deleted files and recover them if the data has not been overwritten.

8. In the NTFS Settings dialog box, **click** the **Search deleted files and folders checkbox** to direct E3 to search for deleted files and folders in the unallocated space in Partion1.

9. **Click** the **Add the Trash folder to the NTFS root checkbox** to direct E3 to add the Trash folder to Partition1 in the Case Content browser.

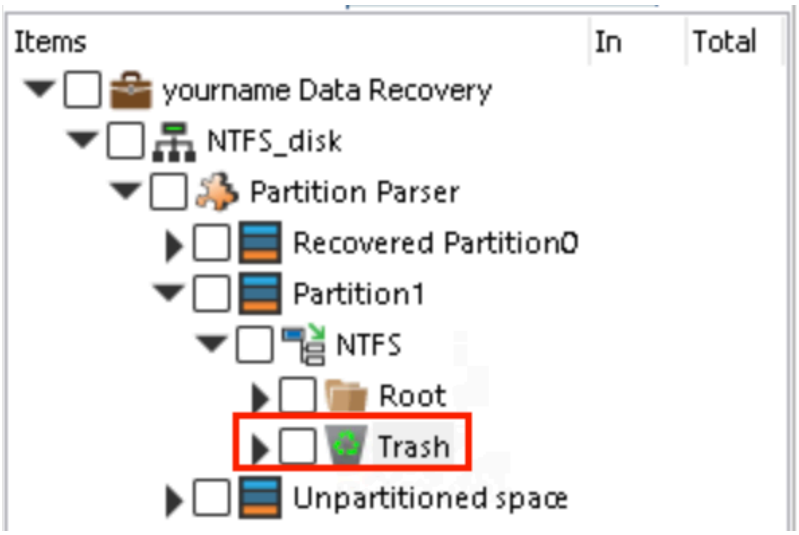
**Note:** In E3, “Trash” refers to the folder where E3 collects any deleted files and folders that it was able to recover from the unallocated space on the drive. It should not be confused with the file system's Trash or Recycle Bin directory, which is still available within the drive image's normal file structure.

10. **Click OK** to close the NTFS Settings dialog box.




NTFS Settings

11. In the Case Content pane, **navigate** to **Partion1/NTFS/Trash** to display the contents of the Trash folder in the Data Viewer.




Trash folder

**Note:** In E3, when a file is identified as deleted in the file system, a red X will appear next to the file name. E3 will attempt to recover the file if possible.

12. In the Data Viewer, **review** the **files recovered by E3.**

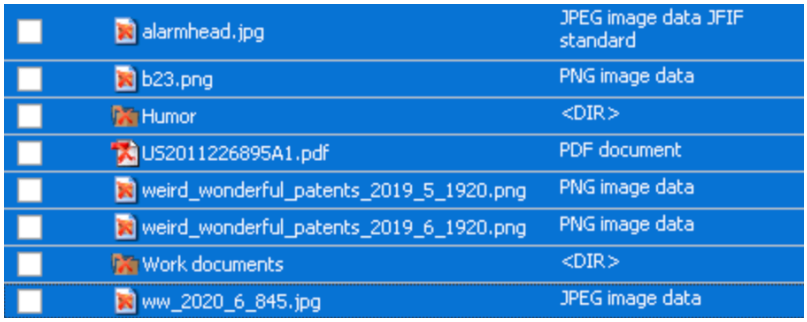
**Note:** Recall that the setting for this lab is an investigation of intellectual property theft. As you select each recovered file in the Trash folder, E3 will display the file in different viewers in the Viewers pane. You may need to expand this pane to see the name of each viewer tab, which includes:

- Properties
- Thumbnails View
- File View
- Document View
- Text View
- Extracted Text View
- Hex View
- File Slack: Text View
- File Slack: Hex View

Using the File Viewer tab, you should see that several of the deleted files contain copies of patents and product designs. Given the scope of the investigation, these files all appear to be potentially incriminating evidence.

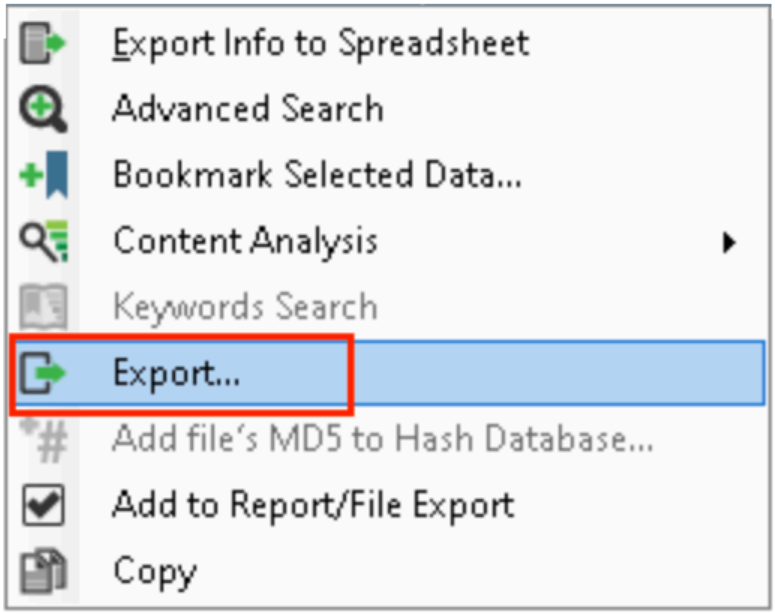
13. **Make a screen capture** showing the **list of recovered files and folders in the E3 Trash folder**.

14. In the Data Viewer, **click** the **alarmhead.jpg file**, then **click** the **ww_2020_ 6_845.jpg file** while **holding the Shift key** to select all of the files.



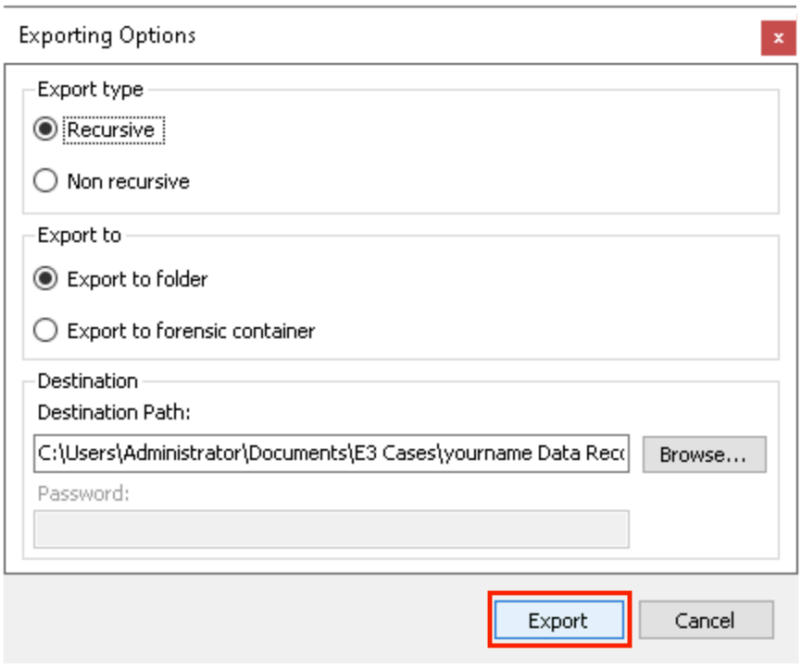

Select the files

15. **Right-click any** of the **selected files**, then **select Export** from the context menu to open the Export Options dialog box.



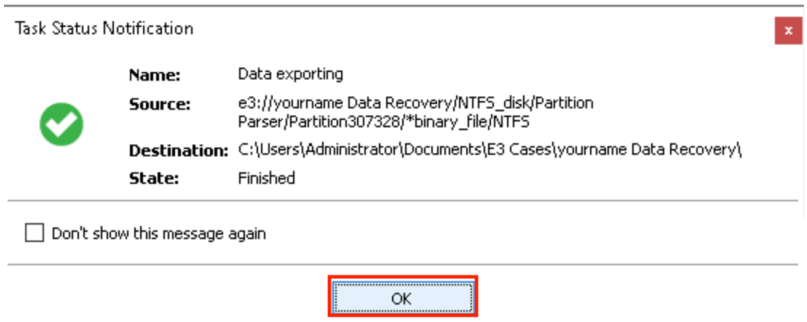

Context menu

16. In the Export Options dialog box, **click Export** to save the recovered files to the case file folder.




Export Options dialog box

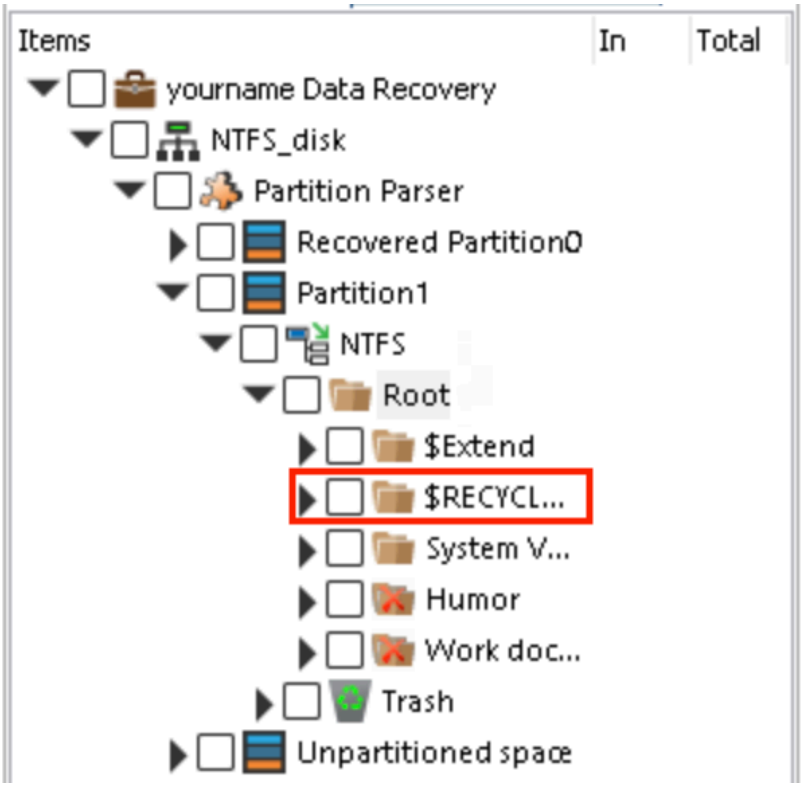
17. When prompted, **click OK** to close the Task Status Notification dialog box.




Task Status Notification dialog box

**Note:** In the next steps, you will extend your investigation to the Windows drive image's Recycle Bin. While the contents of the Recycle Bin are even farther from true deletion than the deleted files you just recovered, many users may not realize this or have simply forgotten to empty the Recycle Bin.

18. In the Case Content pane, **expand** the **Root node**, then **select** the **$RECYCLE.BIN folder** to open it in the Data Viewer.




&RECYCLE.BIN folder

**Note:** When files are moved to the Recycle Bin in Windows 10, two files are created in the Recycle Bin. The first file is assigned a random alpha-numeric name beginning with $I. This file contains the original location of the deleted file. The second file is assigned the same random name, but beginning with $R. The second file is the actual deleted file.

19. **Review** the **contents of the $RECYCLE.BIN folder** and locate a patent file related to sharing location information.

20. **Make a screen capture** showing the **patent file in the File Viewer**.

21. **Repeat steps 15–17** to export the patent file to the case file.

22. **Close** the **E3 window**.

23. In the taskbar, **click** the **File Explorer icon** to open a new File Explorer window.



File Explorer icon

24. In the File Explorer window, **navigate** to the **C:\Users\Administrator\Documents\E3 Cases\yourname Data Recovery folder**.

25. **Make a screen capture** showing the **recovered files in the File Explorer**.

26. **Close** the **File Explorer**.

**Part 2: Recover Deleted Files from Ext 4 Drive Image**

In this part of the lab, you will recover deleted files from a Linux drive image. While Linux is less common as a desktop operating system among casual users, it is widely used in enterprise environments, cybersecurity incidents, and digital forensics investigations. Its open-source nature makes it the backbone for a vast array of devices, including web servers, network appliances, Android smartphones, and IoT devices. Because of this, forensic investigators frequently encounter Linux systems in cases involving server breaches, cybercrime investigations, and embedded system analysis. A strong understanding of Linux from a forensic perspective is essential for tracking unauthorized access, recovering lost or deleted data, and analyzing system logs for evidence.

A key aspect of Linux forensic analysis is its file system, and among the various file systems available, Ext4 (Extended File System version 4) is the most widely used. Ext4 introduces several features that impact forensic investigations, such as journaling, which improves data integrity by recording changes before they are committed to disk, and delayed allocation, which optimizes write performance but can complicate file recovery. Additionally, Ext4's inode-based metadata storage enables efficient file organization but may present challenges when recovering fragmented files. These characteristics influence how data is stored, modified, and potentially recovered, making Ext4 analysis a critical skill for forensic professionals.

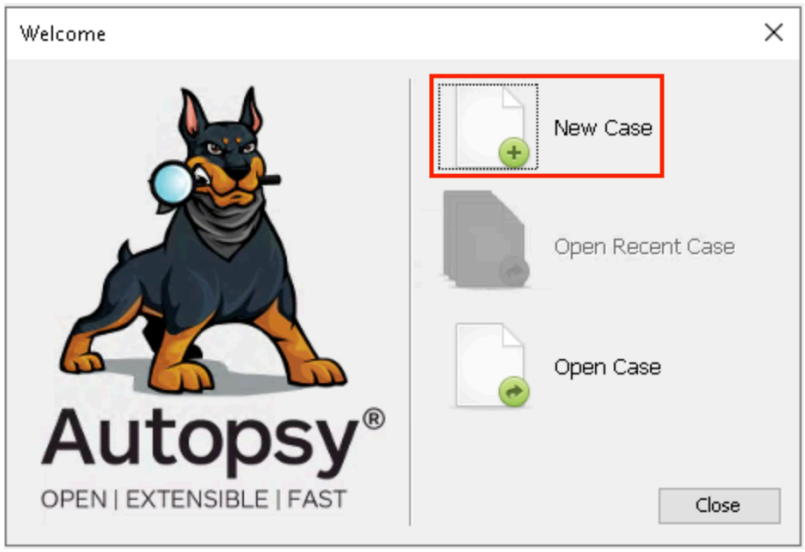
In the next steps, you will use Autopsy, a widely adopted open-source digital forensics tool, to analyze an Ext4-formatted drive image. Once you complete this section, you will have recovered deleted files, interpreted metadata structures, and examined forensic artifacts that are unique to Ext4. By working through this hands-on exercise, you will gain practical experience in file recovery and forensic analysis of Linux file systems, reinforcing your understanding of how forensic methodologies apply to real-world investigations.

1. On the vWorkstation desktop, **double-click** the **Autopsy shortcut** to launch the Autopsy application.



Autopsy icon

2. In the Autopsy Welcome window, **click** the **New Case option** to create a new case file.

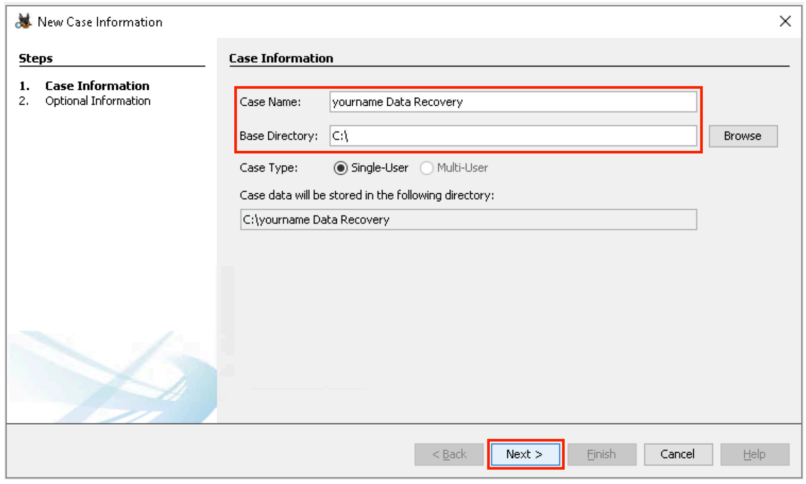



Welcome window

3. On the Case Information page, **type** ***yourname*** **Data Recovery** in the Case Name field.

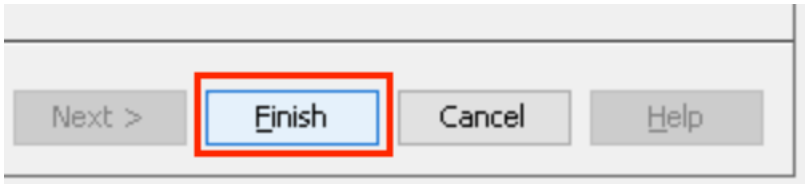
4. **Type** **C:** in the Base Directory field to define the location of the Autopsy case file.

5. **Click Next** to continue to the Optional information page.



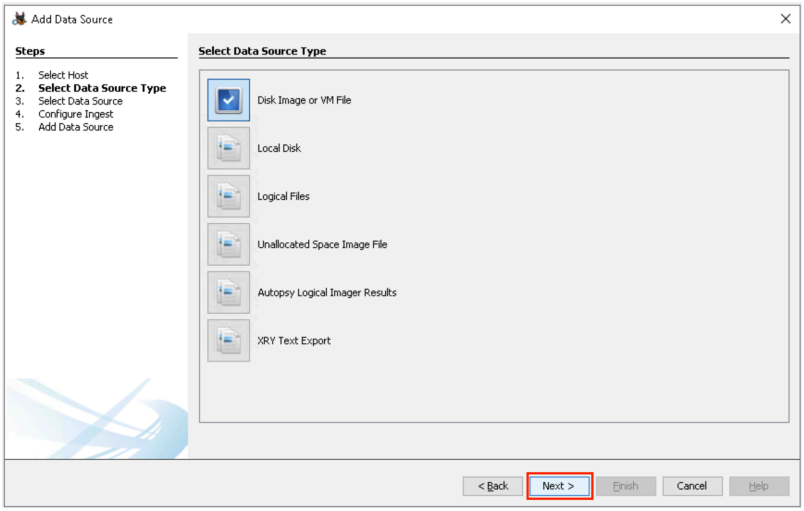

Case Information

6. On the Optional Information page, **click Finish** to save your case file and open the Add Data Source Wizard.



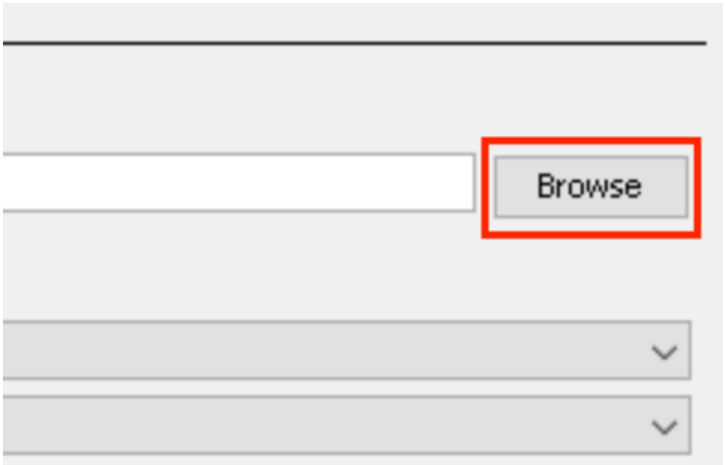

Optional Information

7. On the Select Type of Data Source To Add page, **click Next** to accept the default (Disk Image or VM File) and continue.



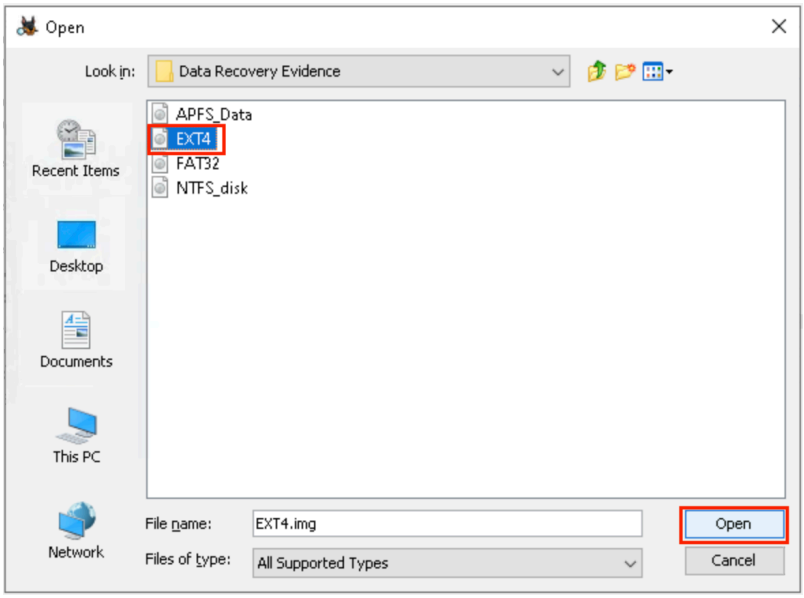

Select Type of Data Source to Add

8. On the Select Data Source page, **click Browse** to open the Browse dialog box.




Browse button

9. In the Browse dialog box, **navigate** to **This PC > Local Disk (C:) > Data Recovery Evidence** and **select** the **EXT4.img file**, then **click Open**.

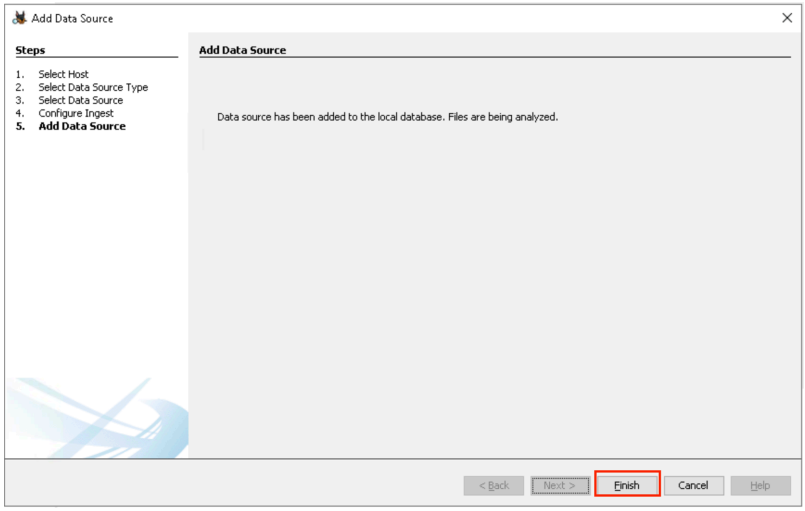



Browse

10. On the Select Data Source page, **click Next** to continue.

11. On the Configure Ingest Modules page, **click Next** to accept the default options and open the Add Data Source page.

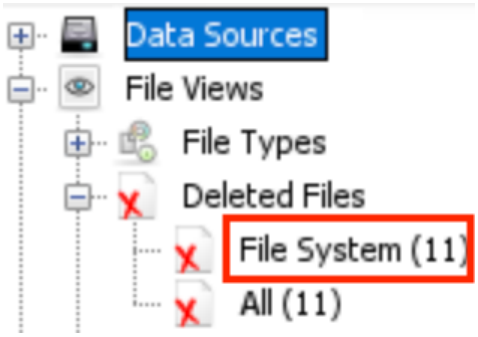
12. On the Add Data Source page, **click Finish** to close the wizard.




Add Data Source

**Note:** Like E3 and NTFS, Autopsy allows you to access deleted data from unallocated space on a drive image. When a drive image is acquired as evidence, Autopsy will automatically scan it for deleted files and attempt to recover them. Any deleted files that Autopsy is able to recover will be moved to the Deleted Files folder in the Tree Viewer, similar to the Trash folder in E3.

13. In the Tree Viewer, **expand** the **Deleted Files node**, then **select** the **File System node** to display the contents in the Result Viewer.

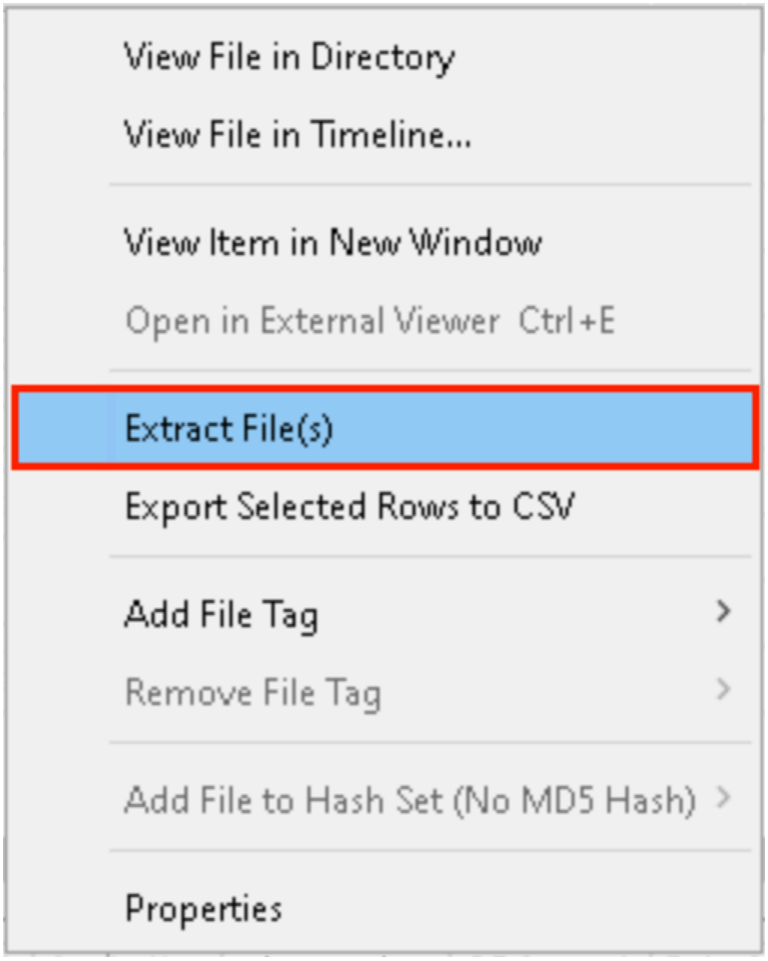


Tree Viewer

14. **Make a screen capture** showing the **contents of the list of deleted files in Autopsy**.

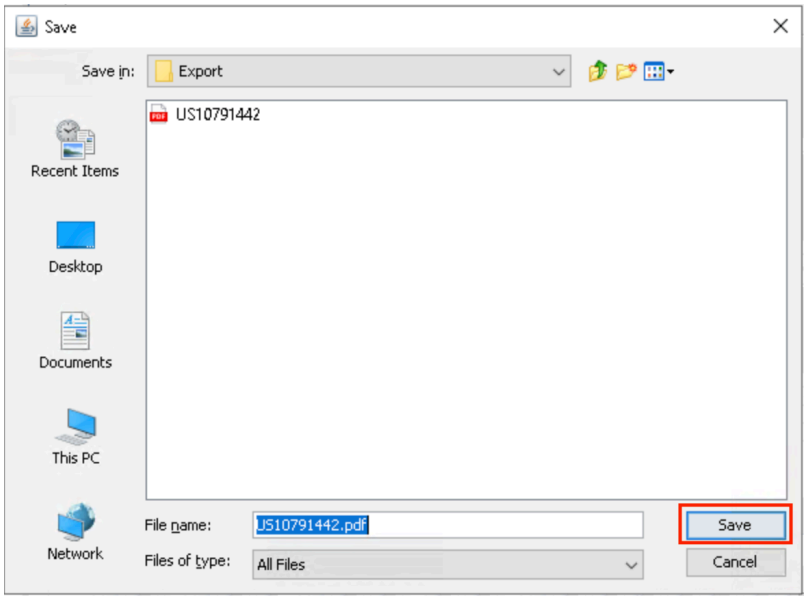
15. **Review** the recovered files in the Result Viewer to locate a patent file.

16. In the Result Viewer, **right-click** the **patent file** and **select Extract File(s)** from the context menu to open the Save window.



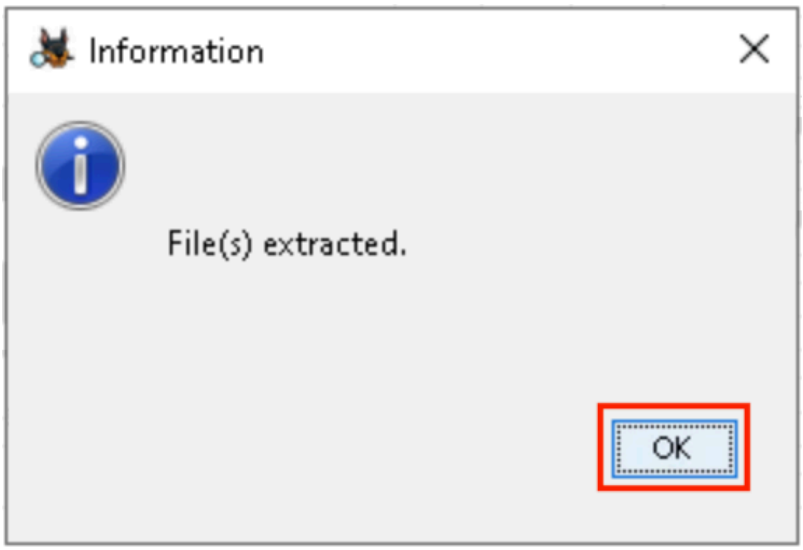

Extract File(s)

17. In the Save dialog box, **click Save** to save the patent file to the C:\yourname Data Recovery\Export folder.




Save window

18. When prompted, **click OK** to dismiss the Information dialog box.




Information dialog box

19. **Close** the **Autopsy window**.

20. From the taskbar, **click** the **File Explorer icon** to open a new File Explorer window.

21. In the File Explorer window, **navigate** to the **C:\yourname Data Recovery\Export folder**.

22. In the Export folder, **double-click** the **patent file** to open it in Adobe Reader.

23. **Make a screen capture** showing the **recovered patent file**.

24. **Close** any **open windows**.

## Applied Learning

### Part 1: Simulating Data Deletion and Corruption

Data carving is a forensic technique used to recover files from raw data on a storage device without relying on the file system's metadata. Unlike traditional file recovery methods that depend on intact metadata structures (e.g., file allocation tables or inodes), data carving reconstructs files based on identifiable patterns within the raw data. This process is particularly valuable in forensic investigations involving accidental formatting, malware

infections, system crashes, or intentional data deletion, where file system corruption renders traditional recovery methods ineffective.

When a file system is corrupted, it may become inaccessible through the usual methods. This is often due to damage to the stored metadata about the file system, which contains information about the system's structure and the location of the various files within the directory tree. Without this information, the data on the drive is just a large jumble of raw unintelligible data. But there are ways to identify the organized chunks of data (files) in these jumbles, so long as you have the right tool.

One common approach to data carving is searching for certain file signatures, such as common headers and footers/trailers for specific file types, that indicate the beginning and ending of a file, which can then be used as cut (or "carve") lines. For example, if you were to inspect any JPEG file with a hex editor (a program that allows you to represent any file in base-16, a.k.a. hexadecimal), you would see the characters FFD8FF at the beginning (header) and the characters FFD9 at the ending (footer/trailer), which may as well say "cut here." Such strings are commonly referred to as "magic numbers," and they provide a way to uniquely identify specific file formats within raw data.

In this part of the lab, you will use PhotoRec, a popular data carving utility, to recover data from a corrupted hard drive. You will first access a Linux file system located on an external drive. You will then delete sensitive data on the drive and perform an intentional corruption of it. After confirming that the device is no longer mountable, you will then use PhotoRec to carve out the data you deleted from the now crippled file system.

1. In the Lab View toolbar, **select TargetLinux01** from the Virtual Machine menu to connect to the TargetLinux01 system.

2. In the TargetLinux01 toolbar, **open** a **new Terminal window**.

3. At the command prompt, **execute** `lsblk -p` to view information about the storage devices.

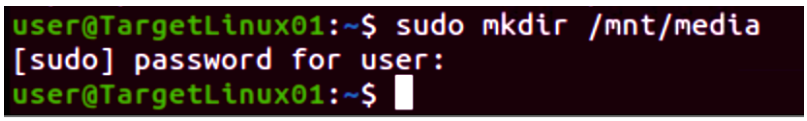
**Note:** The lsblk command lists information about all attached block (storage) devices. Adding the -p option will display the full path to each device discovered, such as /dev/sda1 instead of sda1. The /dev folder is a unique directory that consists mostly of block device files, which store data, and character devices, which typically transfer data. This highlights a unique feature of Linux: everything is considered a file, including attached devices.

Linux follows a convention of labeling the first hard disk sda, the second sdb, and so on, adding numeral suffixes to indicate partitions. You can see that the /dev/sda disk has two partitions mounted (made accessible from a place in your own file tree). The boot partition is mounted at /boot/efi, and the file system is mounted at /. The second disk contains your target ext4 file system, which lives on /dev/sdb2. You will mount this partition in the following steps.

The loop devices can be ignored. These are special files used to mount images in Snap, a package deployment system found in Ubuntu distributions, the mechanics of which are beyond the scope of this lab.

4. At the command prompt, **execute** `sudo mkdir /mnt/media` to create a folder called media within the /mnt folder.

5. When prompted, **type** `password` and **press Enter** to escalate your user privileges.




Create a folder

**Note:** The /mnt directory is commonly used to temporarily mount file systems. You are creating a directory within called media, which is where you will mount the target ext4 file system.

6. At the command prompt, **execute** `sudo mount /dev/sdb2 /mnt/media` to mount the file system.

**Note:** Mounting is made possible as long as that file system's metadata is intact. This file system metadata consists of characteristics such as the size, usage, and location of files and directories on the system. In Linux, this record is known as the superblock.

A successful mount suggests that there is no file system corruption, so you should now be able to access the target ext4 system from within your own file tree.

7. At the command prompt, **execute** `cd /mnt/media/home/ash` to access the user's home directory on the mounted ext4 file system.

8. At the command prompt, **execute** `ls` to display files in the user's directory.

**Note:** You should see a resume and a backup archive. The latter contains some 2FA recovery keys and a server image, as you will find in the next step.

9. At the command prompt, **execute** `rar l backups.rar` to view the contents of the RAR archive.

```
user@TargetLinux01:/mnt/media/home/ash$ rar l backups.rar

RAR 5.50   Copyright (c) 1993-2017 Alexander Roshal   11 Aug 2017
Trial version             Type 'rar -?' for help

Archive: backups.rar
Details: RAR 5

 Attributes      Size     Date    Time   Name
----------- ---------  ---------- -----  ----
 -rw-------         0  2021-08-20 14:19  backups/.recoveryKeys.txt.swp
 -rw-------         0  2021-08-20 14:20  backups/.recoveryKeys.txt.swo
 -rw-r--r--  20000000  2021-08-20 14:22  backups/serverImage.dd
 -rw-r--r--       257  2021-08-20 14:21  backups/recoveryKeys.txt
----------- ---------  ---------- -----  ----
            20000257                    4
```

View backups.rar contents

10. **Make a screen capture** showing the **contents of the RAR archive in the /mnt/media/home/ash directory.**

11. At the command prompt, **execute** `rm -f *` to force (-f) removal of all files (*) in the current directory.

12. At the command prompt, **execute** `cd ~` to return to your home directory so that you are no longer in the mounted ext4 file system.

**Note:** In the next steps, you will deliberately corrupt the file system by writing random raw data to it. This simulates a scenario in which the drive has been damaged such that the superblock is in no shape to facilitate any mounting. Being unable to mount the directory, and without guidance from the file system metadata contained in the superblock, you will need to resort to data carving to recover your files.

13. At the command prompt, **execute** `sudo umount /dev/sdb2` to unmount the drive.

14. At the command prompt, **execute** `sudo dd if=/dev/urandom of=/dev/sdb2 bs=1k seek=200 count=4k` to corrupt the file system.

```
user@TargetLinux01:~$ sudo dd if=/dev/urandom of=/dev/sdb2 bs=1k seek=200 count
=4k
4096+0 records in
4096+0 records out
4194304 bytes (4.2 MB, 4.0 MiB) copied, 0.266969 s, 15.7 MB/s
user@TargetLinux01:~$
```

dd command

**Note:** This command uses a versatile utility called dd, or disk destroyer, to write random data to a section of the file system. The file used to generate input (if=) is /dev/urandom, a character device in the /dev/folder. This urandom device uses the noise produced by operations in the OS to generate pseudo-random numbers and, in your case, is being used to make a complete mess of the partition, which you've designated as the output file (of=/dev/sdb2). You have also specified to write in 1kB blocks (bs=) and to write 4000 (count=4k) of these blocks. Finally, you have specified where to start writing, which is immediately following the first 200 blocks (seek=200).

15. At the command prompt, **execute** `sudo mount /dev/sdb2 /mnt/media` to remount the drive.

**Note:** You should receive an error message indicating that the superblock is damaged ("bad superblock"). The file system metadata cannot be read, so you may attempt to either repair the file system or perform a recovery of certain data on it. Here it is presumed that a file system repair was unsuccessful, so carving is now necessary.

16. **Make a screen capture** showing the **failed mount attempt on the /dev/sdb2 device.**

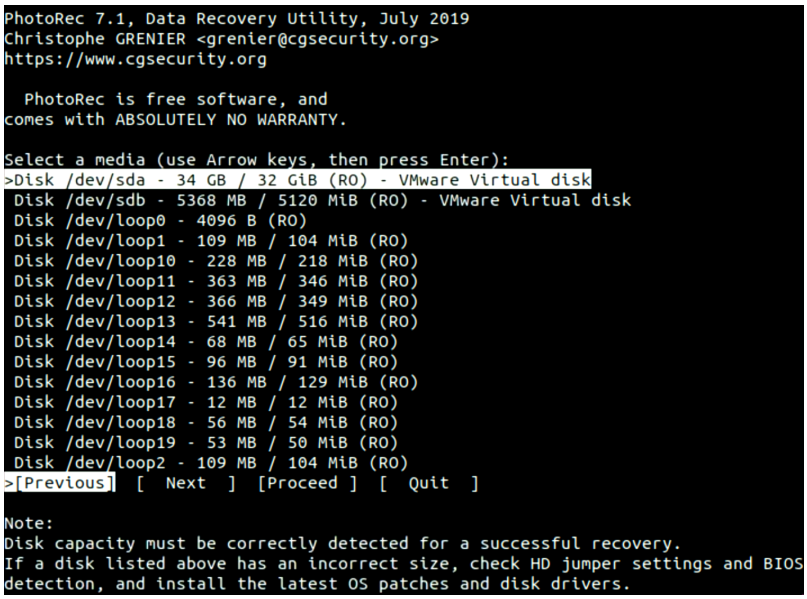
**Part 2: Recovering Deleted and Corrupted Data Using Data Carving**

In this part of the lab, you will use data carving techniques to recover files from the corrupted storage device. Unlike standard file recovery methods, which rely on file system metadata, data carving scans raw storage data to reconstruct deleted files. This approach is crucial in forensic cases where file system structures have been tampered with or destroyed.

The TestDisk package contains two tools, one of which is TestDisk itself. TestDisk can recover both damaged/deleted storage partitions and specific files (so long as that file system is still intact). However, in the event the file system metadata is corrupted and TestDisk is unable to repair the damage itself, its companion utility, PhotoRec, can be invoked to carve the device without such guidance. PhotoRec is a signature-based data-carving tool that ignores the file system and instead reaches for the underlying data directly. This means it is able to recover files even when the file system is severely damaged.

1. At the command prompt, **execute** `sudo apt-get install testdisk` to download and install the TestDisk utility suite.

2. At the command prompt, **execute** `sudo photorec` to launch the data carving application contained within the TestDisk package.



PhotoRec interface

**Note:** Navigating the PhotoRec interface is easy once you understand the layout. All items for the current page are displayed vertically and can be selected using the up/down arrow keys. The bottom menu is traversable using the left/right arrow keys and acts much like the buttons in a typical graphical install wizard (Back, Next, Abort/Cancel, maybe More Options). In this context, the Quit button behaves more like a back button in a web browser, returning you to the previous page in the process unless you are on the first stage (as you are presently), in which case it will exit and terminate the program.

3. You are greeted with an array of drives. Optionally, you could have specified one with the initial command execution.

4. In the Select a media menu, **select** the **/dev/sdb disk**, then **press Enter** to continue.

5. From the Partition menu, **select** the **Linux partition**.

6. From the bottom menu, **select File Opt**, then **press Enter** to continue.

7. At the filetype selection page, **press s** to deselect all file families.

**Note:** PhotoRec can search for around 500 different file types. All are selected by default, but you can select any combination to perform a more targeted search, as you will do in the next steps. Because you are in search of a resume created in Open Office (.odt) and some backups in a RAR archive (.rar), you will select only the file families that contain these file types.

8. Use the arrow keys and spacebar to **select** the **rar** and **zip** file families from the list.

**Note:** Open Office documents are actually compressed files that fall within the zip file family category in PhotoRec.

9. Once both file families are selected, **proceed to** the **next step**.

10. **Press b** to save your selections, then **press Enter** to continue.

11. **Press q** to return to the Partition menu.

12. In the bottom menu, **select Search**, then **press Enter** to continue.

**Note:** The next menu requires you to specify the file system on which your targeted files are stored. While you know the file system on the sdb2 partition is ext4, it is also the most common file system used for Linux, so it is typically a safe bet. Infrequently, you may also see XFS or ReiserFS, in which case you would select the Other option.

13. In the Filesystem menu, **press Enter** to accept the default (ext2/ex3) and continue.

14. In the Space to be analyzed menu, **press Enter** to accept the default (Free) and continue.

**Note:** Electing to scan only the free space is adequate for your purposes. You are interested only in deleted files, so you can expedite your approach by ignoring all allocated space on the disk.

The next menu requires you specify a place to deposit the files. It also admonishes you to not deposit your recovered files to the same media from which you recovered them, which is good advice. To this end, you will use the /home/users/Documents/ directory on your live file system to store your rescues. PhotoRec places you in the same directory you ran the program from, so you will presently find yourself in /home/user, indicated by the Directory line immediately above the file list. You will now select the Documents directory and begin the recovery process.

15. In the Destination menu, **select** the **Documents folder** and **press Enter**, then **press c** to begin carving the file system.

**Note:** As the program processes blocks of data, it will report in real-time the type and quantity of files it has discovered.

This process should take around 3 minutes. Once it has finished, you will see the message “Recovery completed,” after which you may continue to the next step. You will also notice the directory where the recovered data has been saved. You will navigate there in the following steps in search of your compressed files: the .odt resume and the .rar backup.

16. On the Recovery completed page, **press q twice** to exit the PhotoRec program and return to the command prompt.

17. At the command prompt, **execute** `cd Documents/recup_dir.1` to navigate to the folder containing all data recovered by PhotoRec.

18. At the command prompt, **execute** `ls` to display all recovered files.

19. **Make a screen capture** showing the **compressed files recovered by PhotoRec**.

**Note:** Although the names have not been recovered (one of many casualties of the corruption), it appears that the target files have. For some reassurance, you will extract the data from the RAR archive and list the contents in the console to finish out this section.

20. At the command prompt, **execute** `sudo rar e nameofrar` to extract the contents of the RAR archive.

21. At the command prompt, **execute** `ls` to display the recovered backup files.

**Note:** During a real-world recovery operation, you would want to conduct further verification of the files to completely ensure their integrity. For example, you could compare these recovered files against metrics taken from the originals, such as their length or maybe a hash generated beforehand. However, for the purposes of this lab, you can consider this mission successful.

22. **Make a screen capture** showing the **backup files recovered from the RAR archive**.

## Challenge and Analysis

**Note:** The following scenario is provided to allow independent, unguided work, similar to what you will encounter in a real situation.

### Part 1: Recovering Deleted Files from a FAT32 Drive Image

Now that you've had practice recovering deleted files directly from live systems, you will resume the investigation of intellectual property theft that began in Section 1. In the first phase of your investigation, you successfully recovered multiple deleted files and identified potentially incriminating evidence on NTFS and Ext4 file systems. These previous file recovery efforts leveraged key features such as NTFS's MFT and Ext4's journaling capabilities, which helped reconstruct deleted data with relative ease.

This morning, the remaining two drive images were transferred to you for further examination. Your supervisor has asked you to apply the knowledge gained from prior recovery efforts to analyze these new images and uncover additional evidence.

The first drive image is from a Windows system but utilizes the older FAT32 file system. File Allocation Table (FAT) traces its roots back to Windows 95, where it replaced the even older FAT16. While FAT32 has long since been superseded by NTFS on modern Windows desktops, servers, and laptops—primarily due to its lack of robust permission controls—it remains prevalent as the default file system for flash drives and external storage due to its broad compatibility across operating systems and embedded devices.

From a forensic perspective, FAT32 presents unique challenges compared to NTFS and Ext4. Unlike NTFS, which maintains detailed metadata and journaling capabilities that facilitate file recovery, FAT32 lacks journaling and stores only minimal metadata. Deleted files are merely marked as available space in the File Allocation Table without any record of their previous structure or changes, making recovery efforts more complex. Additionally, fragmentation in FAT32 can significantly impact the integrity of retrieved files, necessitating specialized techniques to reconstruct lost data.

Using E3, import the FAT32.img drive image located in the Data Recovery Evidence folder and attempt to recover at least one additional patent file. While doing so, consider how the absence of journaling and weaker metadata structures in FAT32 might affect your recovery approach compared to previous efforts with NTFS and Ext4. Document your findings, noting any challenges encountered and how they differ from your previous recoveries.

1. **Document** your efforts in your forensic report by making a screen capture showing the patent file recovered from the FAT32 drive image within E3.

**Part 2: Recovering Deleted Files from an APFS Drive Image**

Using Autopsy, import the APFS.img drive image located in the Data Recovery Evidence folder and attempt to recover at least one additional patent file.

The second additional drive image was obtained from a computer running Mac OS, which is relevant to this intellectual property theft investigation due to the widespread use of Mac systems in corporate environments, particularly in creative and research-intensive industries. Mac OS is based on the UNIX operating system, sharing core similarities with Linux but incorporating proprietary elements, including its file system.

Historically, Apple computers utilized the Hierarchical File System (HFS) from 1985 until 1998, when Apple transitioned to HFS Plus (HFS Extended) with the release of Mac OS 8.1. In 2017, Apple introduced the Apple File System (APFS) with Mac OS 10.12.4 (High Sierra), replacing HFS Plus. Unlike its predecessors, which were initially designed for mechanical storage devices such as floppy disks and traditional hard drives, APFS is optimized for solid-state drives (SSDs) and incorporates several advanced features, including native encryption, snapshots, space sharing, and improved data integrity.

From a forensic perspective, APFS presents both challenges and advantages. Its built-in encryption, while enhancing security, can significantly hinder forensic access to stored data unless the encryption keys or user credentials are obtained. The snapshot feature allows the system to preserve previous states of the file system, which can either aid forensic recovery by restoring historical data or complicate investigations if unauthorized modifications have been made and preserved in snapshots. Additionally, the space-sharing mechanism complicates traditional file recovery techniques by dynamically allocating storage across multiple volumes, potentially obscuring deleted or overwritten files.

Forensic analysts must use specialized tools capable of parsing APFS structures, handling encryption layers, and interpreting snapshots effectively. In this exercise, you will utilize Autopsy to import the APFS.img drive image located in the Data Recovery Evidence folder and attempt to recover at least one additional patent file. While doing so, consider the potential impact of APFS-specific features on your recovery attempts.

1. **Document** your efforts in your forensic report by making a screen capture showing the patent file recovered from the APFS drive image within Autopsy.